Pre: 9th
Subject:- Biology.
Zain Ali

Based on National Curriculum of Pakistan 2022-23

Biology

Experimentation Skills

Grades 9 & 10

Supplementary Reading Material

National Curriculum Council
Ministry of Federal Education and Professional Training

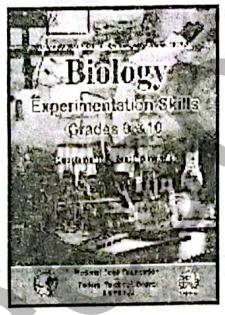
National Book Foundation
as
Federal Textbook Board
Islamabad

Studybeesofficial

CamScanner

© 2024 National Book Foundation as Federal Textbook Board, Islamabad
All rights reserved. This volume may not be reproduced in whole
or in part in any form (abridged, photo copy, electronic etc.)
without prior written permission from National Book Foundation

Biology Experimentat on Skills
for Grades 9 & 10



Author
Prof. Ruqayya Shaikh

TEST EDITION

Management
National Book Foundation

First Edition - First Impression: April 2024 | **Pages:** 50 | **Quantity:** 105000
Printer: Ali Ayyan Printers, Lahore

Note: All the pictures, paintings and sketches used in this book are only for educational and promotional purpose in public interest.

for Information about other publications of National Book Foundation,
visit our Web Site: www.nbf.org.pk or Phone: 051-9261125
or E-mail: books@nbf.org.pk
to share feedback or correction, please send us an email to nbftextbooks@gmail.com

Studybee Official

CamScanner

PREFACE

Experimentation Skills has been developed by NBF according to the National Curriculum of Pakistan 2022- 2023. The aim of this experimentation skills is to enhance learning abilities through inculcation of logical thinking in learners, and to develop higher order thinking processes by systematically building upon the foundation of learning from the previous grades. A key emphasis of the present experimentation skills is on creating real life linkages of the concepts and methods introduced. This approach was devised with the intent of enabling students to solve daily life problems as they go up the learning curve and for them to fully grasp the conceptual basis that will be built upon in subsequent grades.

After amalgamation of the efforts of experts and experienced author, this book was reviewed and finalized after extensive reviews by professional educationists. Efforts were made to make the contents student friendly and to develop the concepts in interesting ways.

The National Book Foundation is always striving for improvement in the quality of its books. The present book features an improved design, better illustration and interesting activities relating to real life to make it attractive for young learners. However, there is always room for improvement and the suggestions and feedback of students, teachers and the community are most welcome for further enriching the subsequent editions of this book.

May Allah guide and help us (Ameen).

Dr. Raja Mazhar Hameed

Managing Director

Studybeesofficial

CamScanner

بِسۡمِ اللّٰہِ الرَّحۡمٰنِ الرَّحِیۡمِ

اللہ کے نام سے شروع جو بڑا مہربان، نہایت رحم والا ہے

Contents

S.NO	Activity	P. No.
1	To study factors affecting rate of diffusion	8
2	To study osmosis using potato strips	10
3	Study of cells and use of a microscope	12
4	Study of animal tissues	15
5	Study of plant tissues	17
6	Qualitative tests for different types of energy foods	21
7	To study the rate of enzyme catalysed reactions	23
8	To study the use of hydrogen carbonate indicator	25
9	To study the rate of photosynthesis	27
10	To study the effect of minerals ions on the growth of plants	29
11	To investigate that exhaled air contains Carbon dioxide	31
12	To study the effect of exercise on heart rate and breathing rate	33
13	To study the reaction time response	35
14	To study the effect of minerals ions on the growth of plants	37
15	To study the process of transpiration in plants	39
16	To study the tropic responses in plants	41
17	To dissect the flowers	43
18	To study the germination of seeds	45
19	To study continuous and discontinuous variation	47
20	To study the ecosystem by using sampling technique	49

Studybeesofficial

CamScanner

LIST OF SLOs

[SLO: B-09-10-X-01] Students should be able to simple measurements in SI Units of:

- volumes of gases or solutions/liquids
- masses
- temperatures
- times
- lengths

[SLO: B-09-10-X-02] Students should be able to carry out simple experiments of:

- diffusion
- osmosis
- food tests
- rates of enzyme-catalysed reactions
- pH and the use of hydrogen carbonate indicator, litmus and universal indicator
- photosynthesis (rate and limiting factors)
- effect of mineral ions on plant growth
- transpiration
- heart rate and breathing rate
- respiration
- tropic responses
- nervous responses
- observation and dissection of seeds and flowers
- germination
- continuous and discontinuous variation
- sampling techniques

[SLO: B-09-10-X-03] Should be able to use of a microscope to examine biological specimens

[SLO: B-09-10-X-04] Calculating the magnification of biological specimens

[SLO: B-09-10-X-05] Students should be able to:

- select and safely use techniques, apparatus and materials
- identify apparatus from diagrams or descriptions
- draw, complete or label diagrams of apparatus and biological specimens
- use, or explain the use of common techniques, apparatus and materials
- select the most appropriate apparatus or method for the task and justify the choice made
- describe food tests
- describe tests to determine the pH of solutions and substances using a universal indicator
- describe and explain techniques
- describe and explain hazards and identify safety precautions
- to ensure the accuracy of observations and data

studybeesofficial

CamScanner

[SLO: B-09-10-X-06] Students should be able to understand for:

- safety measurements and precautions
- understand the need to wear PPE
- tie up long hair
- wear goggles when dealing with caustic materials

[SLO: B-09-10-X-07] Students are able to Understand and express scientific ideas using the below terms:

-True value: the value that would be obtained in an ideal measurement

-Measurement error the difference between a measured value and the true value of a quantity

-Accuracy: a measurement result is described as accurate if it is close to the true value

-Precision, how close the measured values of a quantity are to each other

-Repeatability a measurement is repeatable if the same or similar result is obtained when the measurement is repeated under the same conditions, using the same method, within the same experiment

- Reproducibility: a measurement is reproducible If the same or similar result is obtained when the measurement is made under either different conditions or by a different method or in a different experiment

- Validity of experimental design: an experiment is valid if the experiment tests what it says it will test. The experiment must be a fair test where only the independent variable and dependent variable may change, and controlled variables are kept constant

-Range: the maximum and minimum value of the independent or dependent variables

- Anomaly: an anomaly is a value in a set of results that appears to be outside the general pattern of the results, i.e. an extreme value that is either very high or very low in comparison to others

-Independent variables: independent variables are the variables that are changed in a scientific experiment by the scientist. Changing an independent variable may cause a change in the dependent variable

-Dependent variables, dependent variables are the variables that are observed or measured in a scientific experiment. Dependent variables may change based on changes made to the independent variables

[SLO: B-09-10-X-08] Students are able to:

a. Identify the independent variable and dependent variable

b. describe how and explain why variables should be controlled

c. suggest an appropriate number and range of values for the independent variable

d. suggest the most appropriate apparatus or technique and justify the choice made

Studybees-official

CamScanner

e. describe experimental procedures

f. identify risks and suggest appropriate safety precautions

g. describe how to record the results of an experiment

h. describe how to process the results of an experiment to form a conclusion or to evaluate a prediction make reasoned predictions of expected results

– take readings from apparatus (analogue and digital) or from diagrams of apparatus

– take readings with appropriate precision, reading to the nearest half-scale division where required

– correct for zero errors where required

– make observations, measurements or estimates that are in agreement with expected results or values

– take sufficient observations or measurements

– repeat observations or measurements where appropriate

– record qualitative observations from tests

– record observations and measurements systematically, for example in a suitable table, to an appropriate degree of precision and using appropriate units

– process data, including for use in further calculations or for graph plotting, using a calculator as appropriate

– present data graphically, including the use of best-fit lines where appropriate

– analyse and interpret observations and data, including data presented graphically

– use interpolation and extrapolation graphically to determine a gradient or intercept

– form conclusions justified by reference to observations and data and with appropriate explanation

– evaluate the quality of observations and data, identifying any anomalous results and taking appropriate action

– comment on and explain whether results are equal within the limits of experimental accuracy (assumed to be + 10% at this level of study)

– evaluate experimental arrangements, methods and techniques, including the control of variables

– identify sources of error, including measurement error, random error and systematic error

– identify possible causes of uncertainty in data or in a conclusion

– suggest possible improvements to the apparatus, experimental arrangements, methods or techniques

Studypeesofficial

CamScanner

Activity 1: To study factors affecting rate of diffusion

Materials required	Main skills practised
Glass beakers Tea bags Hot and cold water Watch $KMnO_4$ crystals Stirrer	Planning experiments Performing experiments Following safety precautions Making and recording observations Interpreting results Making conclusions

Introduction

In this activity you will have an opportunity to observe process of diffusion and study which factors affect rate of diffusion.

Points for pre-lab discussion
Q.1. What is diffusion?
Q.2. What are different states in which diffusion can take place?
Q.3. Give examples of diffusion in different states like gas in gas, gas in liquid, liquid in liquid etc.
Q.4. Can we increase the rate of diffusion?

Procedure I

- You are provided with two beakers, one with cold water and other with hot water.
- Add one tea bag in each beaker carefully.
- Do not stir the water.
- Observe the colour change in each beaker after 20 seconds.
- Fill in the following table using the results of your observations.

Time	Colour in cold water	Colour in hot water
20 sec		
40 sec		
60 sec		
80 sec		

Post Lab Discussion

Q.1. Which beaker has darker colour and why?

Q.2. At the start of experiment can you identify the area of high concentration and area of low concentration?

Q.3. Which safety precaution is required in this experiment?

Q.4. What will happen if the contents of beaker are stirred continuously?

Procedure II

- You are provided with two beakers filled with water.

Studybeesofficial

CamScanner

- Add 5 g of $KMnO_4$ crystals in one beaker.
- Add 5 g of $KMnO_4$ powder in one beaker.
- Do not stir the water.
- Observe the colour change in each beaker after 20 seconds.
- Fill in the following table using the results of your observations.

Time	Colour in beaker with crystals	Colour in beaker with powder
20 sec		
40 sec		
60 sec		
80 sec		

Post Lab Discussion

Q.1. Which beaker has darker colour and why?

Q.2. Can you increase the rate of diffusion by some other variable?

Q.3. Identify dependent and independent variable in these experiments.

Studybeesofficial

CamScanner

Activity 2: To study osmosis using potato strips

Materials required	Main skills practised
Potato Salt solutions (0%, 0.5%, 5%, 10%, 20%) Beakers Balance Ruler Blotting paper	Planning experiments Using apparatus Performing experiments Making and recording observations Interpreting results Making graphs

Introduction

In this activity you will have an opportunity to observe process of osmosis in potato strips. You will identify changes in size, mass and turgidity of potato strips after the process of osmosis.

Points for pre-lab discussion
Q.1. What is osmosis?
Q.2. What are different types of osmosis?
Q.3. What are hypertonic, hypotonic and isotonic solutions?
Q.4. Why solutes do not move from area of high concentration to area of low concentration in case of osmosis?

Procedure

- Take 5 beakers and place 0%, 0.5%, 5%, 10% and 20% salt solution in them.
- Label the beakers with marker.
- You have to cut 5 potato strips of equal size. Blot dry.
- Measure the length of strips.
- Measure the mass of strips.
- Try to bend the strips to learn about its turgidity and flexibility.
- Place each of the potato strip in one of the beakers.
- Leave the beakers or 30 minutes.
- Remove each strip, blot dry. Reweigh and measure their length again.
- Check the turgidity and flexibility.
- Fill in the following table using the results of your observations.

Salt solution	Mass at start (g)	Mass at end (g)	Change in mass(end-start) (g)
0%			
0.5%			
5 %			
10%			
20%			

studybeesofficial

CamScanner

Task:

- Make a graph to show the relationship between percentage of salt solution and change in mass.
- Make a table and graph for length of potato strips.

Post Lab Discussion

Q.1. Which salt solutions have increased mass of strips?

Q.2. Which salt solutions have decreased mass of strips?

Q.3. Which salt solutions have increased the turgidity and which ones have increased the flexibility?

Q.4. Which salt solutions are responsible for exosmosis?

Q.5. Which salt solutions are responsible for endosmosis?

Q.6. Can you perform similar experiment by using other materials?

Studybeesofficial

CamScanner

Activity 3: Study of cells and use of a microscope

Materials required	Main skills practised
Compound microscope Glass slides Cover slips Scalpel Forceps Onion skin Tooth pick 1% methylene blue solution Dropper Beaker	Following instructions Using microscope Preparing wet mounts Observing results Making drawings Interpreting results

Introduction

You know that cells are microscopic in size. In this activity you are going to prepare slides to observe some kinds of cells. You will learn about use of microscope and cells.

Points for pre-lab discussion
Q.1. What is a light microscope?
Q.2. If you want to magnify an object 100 times its real size, how can you do this?
Q.3. How does one prepare a wet mount to see cells or other objects under a microscope?
Q.4. How big or small cell is?

Procedure

Part A: Size of an object

- Place a piece of your hair on a glass slide.
- Cover it with cover slip.
- Observe your hair under the microscope with low power objective lens.
- You know the actual size of the hair. It looks much thicker and longer under the microscope.
- Draw a diagram of hair in following table.

Diagram of actual hair	Diagram of hair as seen under microscope

Studybeesofficial

CamScanner

Can you tell how much has it been enlarged?

Calculate the magnifying power of microscope by multiplying magnification of eye piece and objective lens.

Part B: Study of a plant cell

- Place a small piece of onion skin in a drop of water on a slide and cover it with a cover slip.
- Observe it under the microscope first under low power objective then under the high power objective.
- Draw diagrams of onion skin cells in following table.

Diagram under low power objective lens	Diagram under high power objective lens

Part C: Study of an animal cell

- Gently pass the broad end of tooth pick on the inner side of your cheek.
- Place the material o tooth pick in a drop of methylene blue solution on a slide and cover it with a cover slip.
- Observe it under the microscope first under low power objective then under the high power objective.
- Draw diagrams of human cheek cells in following table.

Diagram under low power objective lens	Diagram under high power objective lens

Studybeesofficial

CamScanner

Post Lab Discussion

Complete the following table according to your observation.

Structure	Plant cell	Animal cell
Nucleus		
Cytoplasm		
Cell wall		
Cell membrane		

Q.1. What is the difference between the cells of onion skin and cheek cells?

Q.2. Did you see a cavity (vacuoles) in the plant or animal cell? If so, name the type of cell.

Q.3. What is the colour of onion skin cells? Why they are not green?

Q.4. Which type of cells contain cell wall?

Q.5. Why methylene blue is used?

Q.6. Which cells are larger? Onion skin cells or human cheek cells?

Studybeesofficial

CamScanner

Activity 4: Study of animal tissues

Materials required	Main skills practised
Compound microscope Prepared slides of sections of animal's epithelial, connective, muscle and nervous tissue Charts of animal's epithelial, connective, muscle and nervous tissue	Following instructions Using microscope Observing slides Making drawings Interpreting results Making conclusions

Introduction

In this activity you will have an opportunity to observe animal tissues. You will identify shape of cells and relate it with function of tissue.

Points for pre-lab discussion
Q.1. What is a tissue?
Q.2. How do animals support their body parts, absorb materials or move?
Q.3. Do you except any "photosynthetic tissue" in animals?
Q.4. What is the location of different tissues in animals?

Procedure

- You are provided with charts and slides of four types of tissues.
- Study the features of these tissues from charts.
- Examine each tissue type one by one under the microscope using the low power lens first and then the high power lens to see greater details.
- Fill in the following table using the results of your observations.

TISSUE	DIAGRAM: LABEL MAJOR PARTS
Epithelial tissue	
Connective tissue	

Studybeesofficial

CamScanner

Muscle tissue	
Nervous tissue	

Post Lab Discussion

Complete the following table.

TISSUE	FUNCTION	Feature which compliments functions
Epithelial		
Connective		
Muscular		
Nervous		

Q.1. On the basis of your direct observations and the properties of animal tissues given in above table, can you conclude that the functions of cells are reflected in their structure?

Q.2. Is skin a tissue or an organ? Give reasons to support your answer.

Q.3. What type of tissues are following?

(a) Blood _____

(b) Cartilage _____

(c) Bone _____

(d) Tendon _____

Q.4. How would function of muscle tissue be affected if connective tissue were absent in the body?

Q.5. How would the function of your skin be affected if epithelial tissue is removed from its surface by accident?

Q.6. How would function of the body be affected if nervous tissue/ sensory tissue were absent in your skin?

Studybeesofficial

CamScanner

Activity 5: Study of plant tissues

Materials required	Main skills practised
Compound microscope Charts of sections of plant's leaf, root and stem Prepared slides of sections of plant's leaf, root and stem	Following instructions Using microscope Observing slides Making drawings Interpreting results Making conclusions

Introduction

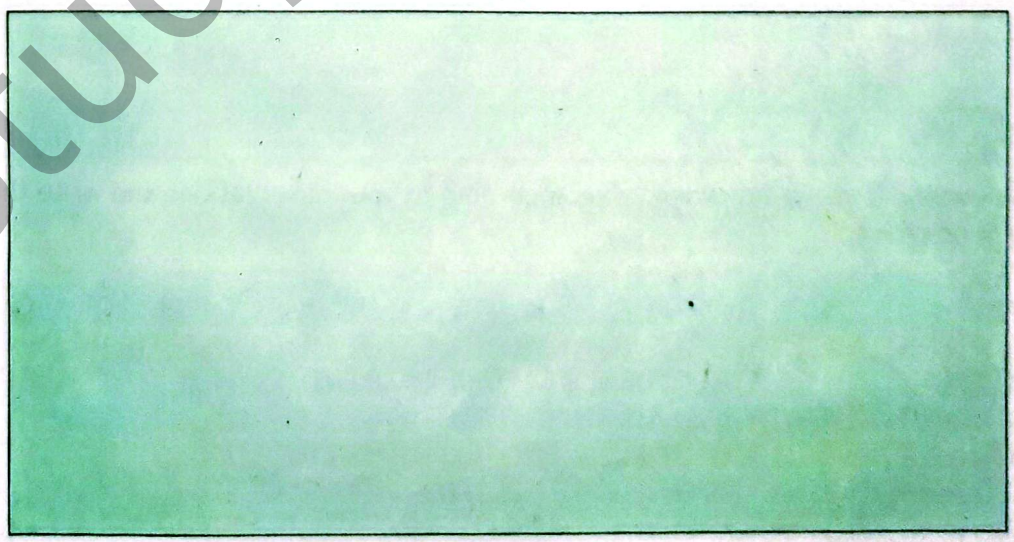
In this activity you will have an opportunity to observe plant tissues. You will identify main tissue types and relate them to their functions.

Points for pre-lab discussion
Q.1. What is meant by permanent slide?
Q.2. How have organisms become multicellular?
Q.3. What is the location of different tissues in plants?

Procedure

Part A: Section of a Leaf

- First take a look at a section of a leaf on a chart.
- Place a prepared slide of a leaf under the microscope using the low power lens first and then the high power lens to see greater details.
- Is the image seen in the microscope inverted as compared to the actual position of the section on the slide?
- Do all cells seen by you in the section look alike?
- Draw a diagram of the section of leaf as seen under your microscope. Label different tissues.



Studybeesofficial

CamScanner

Complete column B of the following table according to your observations and write the name of tissue in column C.

A	B	C
FUNCTION	STRUCTURAL FEATURE OF CELLS	TISSUE
Protection		
Photosynthesis		
Transport		

Q.1. What would be the consequence if the leaf lacked?

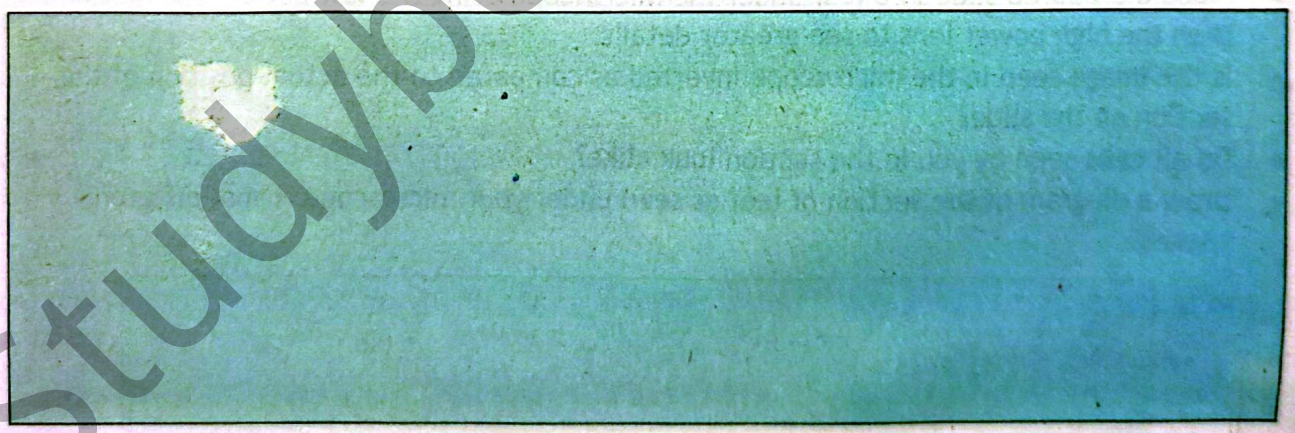
i) Epidermal tissue _____

ii) Vascular tissue _____

iii) Photosynthetic tissue_____

Part B: Section of a Stem

- First take a look at a section of a stem on a chart.
- Place a prepared slide of a stem under the microscope using the low power lens first and then the high power lens to see greater details.
- Draw a diagram of the section of stem as seen under your microscope. Label different tissues.



Complete column B of the following table according to your observations and write the name of tissue in column C.

A	B	C
FUNCTION	STRUCTURAL FEATURE OF CELLS	TISSUE
Protection		
Support		

Studybeesofficial

CamScanner

Transport		
Other (if any)		

Q.1. Do you think that the leaf would get water from the soil if the vascular tissue was absent in the stem?

Part C: Section of a Root

- First take a look at a section of a root on a chart.
- Place a prepared slide of a root under the microscope using the low power lens first and then the high power lens to see greater details.
- Draw a diagram of the section of root as seen under your microscope. Label different tissues.

Post Lab Discussion

Complete the following table.

FUNCTION	TYPES OF TISSUES	FOUND IN		
		LEAF	STEM	ROOT
Absorption				
Protection				
Photosynthesis				
Support				
Conduction				
Storage				

Studybeesofficial

CamScanner

Experimentation Skills

Q1. Which tissues occur in all three organs of plants?

Q2. Is it possible that a tissue may perform more than one function?

Q3. On the basis of your observations, can you give reasons why can the leaf, stem and root be called organs?

Q4. If the leaf, stem and root are called organs, what would you name as organ system in plants?

Q5. Can photosynthesis occur in the leaf if root epidermis and vascular tissue were absent?

Q6. Does the study of tissues suggest to you that multicellular organization of life makes division of labour possible?

Studybeesofficial

CamScanner

Activity 6: Qualitative tests for different types of energy foods

Materials required	Main skills practised
(For a group of four students) 6 test tubes with test tube rack Test tube holder Spirit lamp 3 pipettes, 3 droppers Starch solution 1% Glucose solution 10% Vegetable oil Egg albumin Iodine solution Benedict's solution Ethyl alcohol 95% NaOH solution 10% Copper sulphate solution 2%	Following instructions Using apparatus Observing changes Interpreting results Making conclusions

Introduction

We need food to get energy, growth and repair. In this activity you will learn how to detect various nutrients food items.

Points for pre-lab discussion
Q.1. What nutrients does food contain?
Q.2. What are the unique roles of each of the nutrients in human body?
Q.3. How can we check the presence or absence of nutrients in food in the laboratory?

Procedure

1. Test for starch (Iodine test):
   Take 2ml of 1% starch solution in a test tube with a pipette.
   Add few drops of 1% iodine solution drop by drop with a dropper.
   What happens to the colour of solution? Record it.

   _____

2. Test for Glucose (Benedict's test):
   Take 2ml of a glucose solution in a test tube.
   Add 2 ml of Benedict's solution.
   What is the colour of the solution? Record it.

   _____

   Shake and boil for 2 minutes and allow to stand. Use a test tube holder to hold the tube over the flame.
   Does the colour change? Record it.

   _____

Study bees official

CamScanner

3. Protein test (Biuret test)
   Take 2ml of egg albumin solution in a test tube.
   Add equal volume of 5% sodium hydroxide solution.
   Shake well.
   Add 2 drops of 1% copper sulphate solution and mix.
   What change do you observe? Record it.

4. Test of fatty substances (Emulsion test):
   Add 2ml fat or oil to a test tube containing 2ml of absolute ethanol.
   Dissolve the lipid by shaking vigorously.
   Add equal volume of cold water.
   Shake then allow to stand.
   What change do you observe? Record it.

Post Lab Discussion

Complete the following table using your results.

Name of substance	Name of test	Observation
Starch		
Glucose		
Albumin		
Oil		

The expected colour changes are as follows:

For Starch : deep blue
For Glucose : red precipitate
For protein : purple
For fat : white cloudy emulsion

Q.1. On the basis of your direct observations can you conclude that your results are positive for each food type. If not try to give possible reasons.

Q.2. In the above activity, the nutrients to be tested are already known (e.g., starch, glucose, protein, oil). What do you think is the usefulness of your learning these tests?

Precaution: Wear Lab coat, safety goggles and gloves as NaOH solution, Iodine solution and copper sulphate solution are hazardous.

Studybeesofficial

CamScanner

Activity 7: To study the rate of enzyme catalysed reactions

Materials required	Main skills practised
(For a group of four students) Test tubes with test tube rack Incubator/ water bath Pipettes, droppers Starch solution 1% Iodine solution Benedict's solution Amylase/ diastase solution	Following instructions Using apparatus Observing changes Interpreting results Making conclusions

Introduction

Enzymes are catalysts that speed up metabolic reactions in living things. In this activity you will find rate of enzyme catalysed reaction.

Points for pre-lab discussion
Q.1. What is the chemical nature of enzymes?
Q.2. What are the important sites of enzyme?
Q.3. How can we check the rate of action of enzymes in the laboratory?

Procedure

- Label five test tubes 1-5.
- Place 4 mL of 1 % starch in each of the test tubes.
- Place all of the tubes in the 37°C water bath for 5 minutes.
- Tube 1 will be your control and it will not contain any enzyme.
- Remove the tubes from the water bath momentarily, and quickly add 3 drops of the warmed amylase solution to tubes 2-5.
- Mix the tubes quickly by shaking them gently, and immediately put them back in the 37°C water bath.
- Record the time at which you added enzyme.
- Remove tubes from water bath according to the time given in table.
- Add one small drop of the iodine reagent to each tube.
- Record your observations in the following table.

Number of tube	Time of tube removal	Observation (Colour after Iodine test)
1	15 minutes	
2	2 minute	
3	5 minutes	
4	10 minutes	
5	15 minutes	

Studybeesofficial

CamScanner

Post Lab Discussion

The expected colour changes are as follows:

Iodine test result for starch	Amount of Starch remaining	Enzyme activity level
Dark blue black	All	None
Blue	Most	Low
Brown	Some	Moderate
Light brown	None	High

Q.1. When is the high enzyme activity level achieved?

Q.2. Why tube 1 result is dark blue colour?

Q.3. Can you use your saliva if amylase solution is not available?

Q.4. Why water bath is used in this experiment?

Q.5. Can you modify this experiment to study various factors affecting rate of enzyme reactions?

Precaution: Wear Lab coat, safety goggles and gloves as iodine solution is hazardous.

Studybeesofficial

CamScanner

Activity 8: To study the use of hydrogen carbonate indicator

Materials required	Main skills practised
Test tubes	Following instructions
Test tube rack	Using apparatus
Fresh leaves	Observing changes
Hydrogen Carbonate solution	Interpreting results
Pipette	Making conclusions

Introduction

The hydrogen carbonate indicator can detect increase or decrease in carbon dioxide concentration. It is used to monitor both photosynthesis and respiration.

Points for pre-lab discussion
Q.1. What is indicator?
Q.2. Which process in plants produces carbon dioxide?
Q.4. Which process in plants uses carbon dioxide?

Procedure

- Label the test tube as A, B, and C.
- Put fresh leaves in test tubes B and C in such a way that they are not at the bottom of tube.
- The test tube A is kept without leaf as control tube.
- Put 5 ml of hydrogen carbonate solution in each test tube with the help of pipette.
- Place cork bungs on top of the tubes.
- Cover test tube C with aluminium foil.
- Place all tubes in a rack and put the rack in sun light.
- After 30 minutes observe the colour of the test tubes.
- Record the observations in the table below.

Test tube	Conditions	Initial colour	Final colour
A	No leaf (Control)		
B	Leaf in sunlight		
C	Leaf wrapped in aluminium foil		

Studybeesofficial

CamScanner

Post Lab Discussion

The expected colour changes are as follows:

Tube with no leaf - No change

Tube with leaf in sunlight - Purple colour

Tube with leaf in darkness - Yellow colour

Q.1. Are your findings different from expected colour change? If yes, then repeat experiment with adding two leaves in tube B and C.

Q.2. What the orange colour in tube A indicates?

Q.3. Which gas is more in test tube B and why?

Q.4. Which gas is more in test tube C and why?

Studybeesofficial

CamScanner

Activity 9: To study the rate of photosynthesis

Materials required	Main skills practised
Baking soda Beaker Dish soap Plant leaves (spinach) Single hole puncher Plastic syringe 10 ml (without needle) Timer	Following instructions Planning experiments Using apparatus Observing changes Interpreting results Making conclusions

Introduction

The process of photosynthesis takes place in green parts of plants. In this activity you will observe this process in leaf discs.

Points for pre-lab discussion
Q.1. Define the process of photosynthesis?
Q.2. What are the requirements of the process of photosynthesis?
Q.3. What are the products of photosynthesis?
Q.4. Why the process of photosynthesis is important for life?

Procedure

- Fill a beaker with 300 mL water. Add about 1/8 teaspoon baking soda and 1 drop of liquid dish soap to each beaker. Mix gently.
- Punch out 10 leaf disks from the spinach leaves using the hole puncher.
- Remove the plunger of the syringe and place 10 leaf disks into the syringe barrel.
- Place the plunger back into the syringe and push it down until only a small volume of air is left in the syringe. Be careful not to crush the leaf disks.
- Suck up a small volume of the baking soda solution from one of the cups into the syringe with the leaf disks and hold it vertically. The leaf disks should all float on the surface of the solution.
- Remove the air pockets within the leaves to make the leaf disks sink, as follows.
  - Carefully push out all the air from the syringe.
  - Close the opening of the syringe with a finger and pull back on the plunger to create a vacuum. Hold the vacuum for 10-15 seconds and swirl the leaf disks to suspend them in the solution.
  - Stop pulling on the plunger and remove your finger from the syringe opening to release the vacuum.
- Repeat step 6 until all leaf disks have sunk to the bottom of the solution.
- Place leaf discs in beaker with baking soda solution.
- Place beaker with the leaf disks under your light source.
- Start a timer. Observe closely what happens to the leaf disks.
- At the end of each minute, record the number of floating disks in the given data table, until all of the leaf disks are floating.

Studybeesofficial

CamScanner

Time	Number Of Floating discs	Time	Number Of Floating discs
0 min		6 min	
1 min		7 min	
2 min		8 min	
3 min		9 min	
4 min		10 min	
5 min		11 min	

Make a graph of above data. Plot the time on x-axis and number of floating discs on y-axis.

Rate of photosynthesis graph

Post Lab Discussion

Q.1. Do you think rate of photosynthesis remains same all the time?

Q.2. How can we increase the process of photosynthesis?

Q.3. Which factors affect rate of photosynthesis?

Q.4. Can you study the effect of carbon dioxide concentration on rate of photosynthesis by using this experiment? Discuss your plan.

Q.5. Can you study the effect of different colours of light on rate of photosynthesis by using this experiment?

Q.6. Can you study the effect of intensity of light on rate of photosynthesis by using this experiment?

Q.7. Can you study the effect of temperature on rate of photosynthesis by using this experiment?

Studybeesofficial

CamScanner

Activity 10: To study the effect of limiting factors on rate of photosynthesis

Materials required	Main skills practised
Experimental data Graph paper	Analysing data Plotting graph Interpreting results Making conclusions

Introduction

The process of photosynthesis depends on many factors. In this activity you will study the effect of limiting factors on rate of photosynthesis.

Points for pre-lab discussion
Q.1. Name the factors on which the process of photosynthesis depends?
Q.2. How a factor becomes limiting factor for the process of photosynthesis?

Procedure

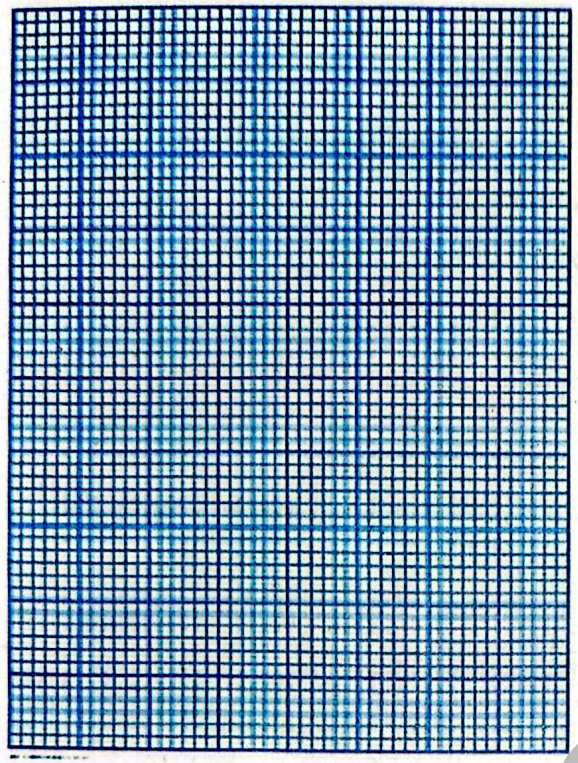
An experiment was carried out to find the rate of photosynthesis at different carbon dioxide concentrations. This was repeated at two different light intensities. The results are given in the following table:

CO_2 Concentration %age	Rate of Photosynthesis in arbitrary units	
	Low light intensity	High light intensity
0.0 %	0	0
0.02 %	20	20
0.04 %	29	35
0.06 %	35	47
0.08 %	39	68
0.10 %	42	84
0.12 %	45	89
0.14 %	46	90
0.16 %	46	90
0.18 %	46	90

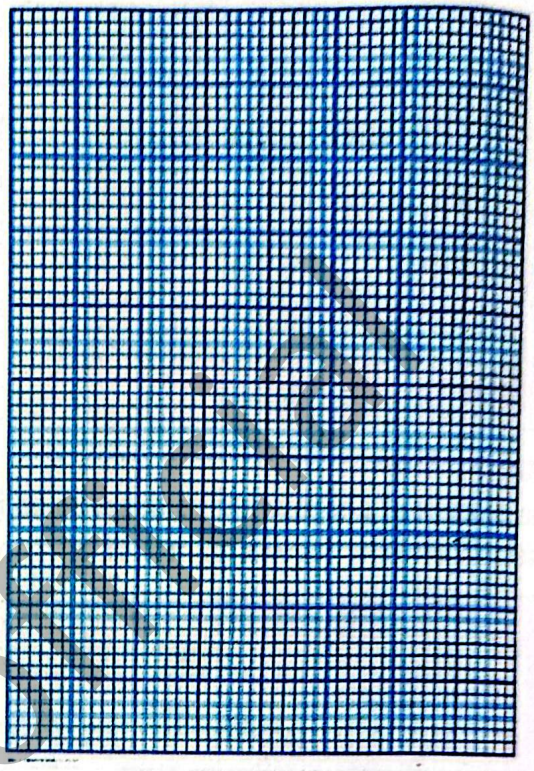
Plot these results on following graphs. Use CO_2 concentration on x-axis and rate of photosynthesis on y-axis.

Studybeesofficial

CamScanner



Low light intensity graph



High light intensity graph

Post Lab Discussion

Q.1. Do you think rate of photosynthesis is same in both graphs?

Q.2. Describe the patterns shown by these graphs.

Q.3. Which regions of these graphs show that carbon dioxide is a limiting factor?

Q.4. How you identified the carbon dioxide limiting factor region in these graphs?

Q.5. Which regions of these graphs show that light intensity is a limiting factor?

Q.6. How you identified the light intensity limiting factor region in these graphs?

Q.7. What other factors can affect the rate of photosynthesis?

Q.8. How can we increase the rate of photosynthesis?

Q.9. Should the farmers know about liming factors of photosynthesis?

Studybeeso

CamScanner

Activity 11: To investigate that exhaled air contains Carbon dioxide

Materials required	Main skills practised
Calcium hydroxide	Following instructions
Rubber tube/ straw	Planning experiments
Flasks	Preparing solution
Filter paper	Making observations
Funnel/ large test tube	Recording results
10 ml Syringe	Making conclusions

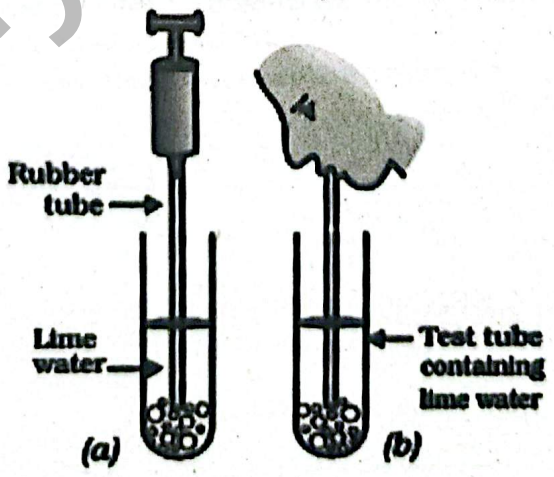
Introduction

During breathing you inhale oxygen and exhale carbon dioxide. You will perform an activity to investigate which gas you exhale.

Points for pre-lab discussion
Q.1. What is inhalation and exhalation?
Q.2. What are the percentage of gases in inhaled and exhaled air?
Q.3. What is the relation between cellular respiration and breathing?

Procedure

- Prepare fresh lime water by using Calcium hydroxide. (Mix a tea spoon of calcium hydroxide in one cup of tap water. Let it stand for 10 minutes. Some white solid will settle down. Filter the top solution by using filter paper and funnel. Use this transparent filtrate as fresh lime water)
- Take some freshly prepared lime water in two flasks/ large test tubes.
- Blow air by mouth in one flassk using straw.
- Blow air by syringe in other flask using rubber tube.
- Note how long it takes to turn lime water milky.



National Book Foundation
as
Federal Textbook Board
Islamabad
Studybeesofficial
CamScanner

Fig: Set up to investigate presence of CO_2 in exhaled air

Post Lab Discussion

Q.1. Which flask's lime water becomes milky first?

Q.2. Why exhaled air contains more carbon dioxide?

Q.3. Which molecule is formed by reaction of lime water and carbon dioxide?

Q.4. Do you know that calcium hydroxide is available at pan shops as 'Chuna'?

Q.5. Why it is important to use freshly prepared lime water?

Q.6. Do you think that plants also release carbon dioxide?

Q.7. Can you plan an experiment to investigate release of carbon dioxide from germinating seeds?

Studybeesofficial

CamScanner

Activity 12: To study the effect of exercise on heart rate and breathing rate

Materials required	Main skills practised
Timer Open space for exercise	Following instructions Planning experiments Observing changes Recording results Making conclusions

Introduction

During exercise your muscles work more so they need more energy. You will perform an activity to compare the rate of heart beat and breathing before and after exercise.

Points for pre-lab discussion
Q.1. What is the range of normal heart rate?
Q.2. Which factors can change heart rate?
Q.3. Is heart rate related with breathing rate?
Q.4. Why muscles need more energy during exercise?

Procedure

- Find your pulse either in your neck or your wrist.
- Count the number of beats you can feel in 30 seconds.
- Multiply it by two to get beats per minute and record it in table as resting heart rate.
- Count the number of breaths in 30 seconds, multiply it by two and record as resting breathing rate.
- Perform some moderate exercise like running, jumping or jogging for four minutes.
- Count heart beats per minute and breaths per minute and record your observations as after exercise rates in the following table.
- Repeat this process three times and calculate average.

Readings	Resting Heart Rate	Heart Rate After Exercise	Resting Breathing Rate	Breathing Rate After Exercise
1st				
2nd				
3rd				
Average				

Studybeesofficial

National Book Foundation
as
Federal Textbook Board

CamScanner

Post Lab Discussion

Q.1. What is the change in heart rate and breathing rate after exercise?

Q.2. Is there any change in breathing pattern before and after exercise?

Q.3. Is there any change in your sweating and face redness before and after exercise?

Q.4. Is your body trying to maintain homeostasis?

Q.5. If you repeat this experiment with measuring blood pressure by using a digital BP apparatus, what will happen?

Q.6. Are your results near true value?

Q.7. If a student records three readings of resting breathing rate as 20, 35 and 10; Identify the type of anomaly/ error in his result.

Q.8. Discuss the findings of your whole class to determine the range of values of heart rate and breathing rate.

Q.9. Which students returned back to normal condition earlier after exercise?

Q.10. After this experiment can you identify which students are physically fit in your class?

Q.11. Identify dependent, independent and controlled variables of this experiment from the following list; timer, duration of exercise, type of exercise, resting heart rate, resting breathing rate, heart rate after exercise, breathing rate after exercise, fitness of student, age of student, weight of student, gender of student, weather conditions.

Studypeesofficial

CamScanner

Activity 13: To study the reaction time response

Materials required	Main skills practised
A pair of students 30 cm ruler	Following instructions Planning experiments Recording results Making conclusions

Introduction

Nervous system controls your responses. You will perform an activity of ruler drop test to find your reaction time response.

Points for pre-lab discussion
Q.1. What are different types of nervous responses?
Q.2. Which stimuli can change your responses?
Q.3. Does everyone respond at the same time?
Q.4. Are your both hands similar in responding to stimuli?

Procedure

- One student stands and holds the ruler near 30 cm mark and lets it hang vertically.
- The other sitting student holds their thumb and index finger of right hand near 0 cm mark, ready to catch it when it falls. This student should not touch the ruler.
- Without warning the student holding the ruler should let go and the other student tries to catch the ruler as quickly as possible.
- The level (in cm) where the ruler was caught is recorded as drop distance.
- The same student is tested 5 times and average of these readings is calculated.
- Repeat above steps for left hand.
- Fill the following table. Use reference table for reaction time in seconds.

Readings	Right hand		Left hand	
	Drop distance (cm)	Response time (s)	Drop distance (cm)	Response time (s)
1st				
2nd				
3rd				
4th				
5th				
Average				

Studybeesofficial

CamScanner

Reference table for drop distance and reaction time.

Drop distance (cm)	Reaction time (s)	Drop distance (cm)	Reaction time (s)	Drop distance (cm)	Reaction time (s)
1	0.04	11	0.15	21	0.21
2	0.06	12	0.16	22	0.22
3	0.08	13	0.16	23	0.22
4	0.09	14	0.17	24	0.22
5	0.10	15	0.18	25	0.23
6	0.11	16	0.18	26	0.23
7	0.12	17	0.19	27	0.24
8	0.13	18	0.19	28	0.24
9	0.14	19	0.20	29	0.24
10	0.14	20	0.21	30	0.25

Post Lab Discussion

Q.1. Which hand is quick in your reaction time response?

Q.2. Can you improve your reaction time response?

Q.3. A student had a sleepless night before experiment. He showed slow responses in this activity. Can you identify the reason?

Q.4. Is there any food or drink which can improve this response? Give reasons.

Q.5. In this activity you have used your eyes as receptor. Plan an activity to use sound instead of vision.

Q.6. Why five readings are taken? What happens when only one reading is taken?

Q.7. A student repeats this activity for 10 days. Will there be any change in the results after 10 days?

Q.7. Identify dependent, independent and controlled variables of this experiment from the following list; ruler, alertness of student, dominant hand of student, eyesight of student, reference table, behaviour of partner, health conditions of student, food habits of student.

Studybeesofficial

CamScanner

Activity 14: To study the effect of minerals ions on the growth of plants

Materials required	Main skills practised
Young wheat or pea seedlings Test tubes Test tube rack Mineral solution of urea Mineral solution of phosphate rock Mineral solution of potassium chloride Washed sand Balance Ruler	Following instructions Planning experiments Using apparatus Observing changes Interpreting results Making conclusions

Introduction

The normal growth of plants requires minerals specially nitrogen, phosphorous and potassium. In this activity you will learn about role of some minerals in plant growth.

Points for pre-lab discussion
Q.1. What is the importance of nitrogen for plant metabolism?
Q.2. Why plants need phosphorous and potassium?
Q. 3. What other minerals are required by plants and why?

Procedure

- Prepare dilute solutions of minerals (Use urea for nitrogen, rock phosphate for phosphorous and potassium chloride for potassium).
- Label 5 tubes from 1-5.
- Fill all tubes with washed and dried sand.
- Plant same seedlings in all tubes.
- Add distilled water to test tube 1.
- Add nitrogen solution to test tube 2.
- Add phosphorous solution to test tube 3.
- Add potassium solution to test tube 4.
- Add a mixture of all mineral solutions to test tube 5.
- Cover all test tubes with black paper and place in the test tube rack.
- Regularly provide same solution in required quantity. Make sure the sand is moist all the time.
- After two weeks take out seedlings and measure the root length, shoot length and mass. Also count the number of leaves and note the colour of leaves.
- Record your observations in the following table.

studybeesofficial

CamScanner

Test tube	Conditions	Shoot length (cm)	Root length (cm)	Mass (g)	Number of leaves	Colour of leaves
1	Without nutrients					
2	With nitrogen					
3	With phosphorous					
4	With potassium					
5	With nitrogen, phosphorous and potassium					

Post Lab Discussion

Q.1. Which tube has maximum growth and why?

Q.2. Which tube has minimum growth and why?

Q.3. Do you think that plant growth can be further increased?

Q.4. How seedlings grow in test tube 1?

Q.5. Why leaves have turned yellow in some tubes?

Q.6. Why some plants have stunted growth?

Studybeesofficial

CamScanner

Activity 15: To study the process of transpiration in plants

Materials required	Main skills practised
Cobalt chloride paper Scotch tape Scissors Plant with flat green leaves (orange, geranium etc.)	Following instructions Planning experiments Using apparatus Observing changes Interpreting results Making conclusions

Introduction

Transpiration is the evaporation of water from plants. The water is lost in the form of water vapours into the atmosphere through the stomata. You will perform an activity to compare the rate of transpiration between upper and lower epidermis.

Points for pre-lab discussion
Q.1. What is the structure of leaf?
Q.2. When does transpiration take place?
Q.3. Why plants need to open their stomata?
Q.4. What is the importance of transpiration?

Procedure

- Prepare solutions of cobalt chloride and soak filter papers in it.
- Dry the soaked papers thoroughly.
- Cut small pieces of cobalt chloride paper.
- Select a green leaf of plant in sunlight.
- Attach a piece of cobalt chloride paper on upper and lower epidermis of leaf.
- Properly cover cobalt chloride paper with scotch tape.
- Observe each paper after every 5 minutes.
- Note the colour change and record your observations in the following table.

Time (min)	Upper epidermis paper colour	Lower epidermis paper colour
0		
2		
4		
6		
8		
10		

Post Lab Discussion

Q.1. What is the colour of dry cobalt chloride paper?

Q.2. What is the colour of cobalt chloride paper at the end of experiment?

Federal Textbook Board
Islamabad

Studybesofficial

CamScanner

Q.3. Which surface of epidermis changes the colour of cobalt chloride paper first and why?

Q.4. Can you use paper tape instead of scotch tape?

Q.5. Why bifacial leaves are used in this experiment?

Q.6. If you repeat this experiment with mono facial leaves like lily leaves, what will happen?

Q.7. Will you get similar results in summer and winter? Give reason.

Q.8. Will there be any effect of humidity percentage on this experiment's findings?

Q.9. Can you perform this activity at night? Give reason to support your answer.

Studybeesofficial

CamScanner

Activity 16: To study the tropic responses in plants

Materials required	Main skills practised
A pair of petri dishes Soaked corn grains Blotting paper or filter paper Cotton Water proof ink marker Scotch tape Stand with clamp	Following instructions Planning experiments Using apparatus Observing changes Interpreting results Making conclusions Drawing diagrams

Introduction

Tropic responses are the growth movements in plants due to external stimuli. The external stimuli may be light, gravity, water, contact and chemicals. In this activity you will study tropic response of seedling parts to gravity.

Points for pre-lab discussion
Q.1. What are some examples of plant movements?
Q.2. Can plants show quick responses/ movements?
Q.3. What are positive tropic movements?
Q.4. What are negative tropic movements?

Procedure

- Take a petri dish. Draw two intersecting lines, at the bottom of the petri dish, passing through the centre of the dish.
- Take four soaked corn grains, and place one on each line with their pointed ends directed towards the centre.
- Cut a piece of blotting or filter paper to fit lightly inside the bottom of the petri dish.
- Cover the corn grains with that paper.
- Pack rest of the dish tightly with wet cotton.
- Close the dish and seal its edge with the strips of scotch tape.
- Make the dish to stand on its edge with the help of clay or stand with clamps.
- Leave this setup in the laboratory for 4 days.
- After 4 days, make and record your observations.

In which direction all roots are growing?

_____

In which direction all the shoots are growing?

_____

Study beesofficial

CamScanner

Draw the petri dish with seeds in the following space.

Petri dish with seeds at the start	Petri dish with seeds after 4 days

Post Lab Discussion

Q.1. What is geotropism?

Q.2. Which part of plant shows negative geotropism?

Q.3. Which part of plant shows positive geotropism?

Q.4. What can be the benefit of geotropism to the plants?

Q.5. What is phototropism?

Q.6. Can you plan an experiment to investigate phototropism?

Q.7. What can be the benefit of phototropism to the plants?

Studybeesofficial

CamScanner

Activity 17: To dissect the flowers

Materials required	Main skills practised
Fresh Flowers	Following instructions
Paper plates	Observing parts
Forceps	Interpreting results
Scissors	Making comparisons
Magnifying lens	Drawing diagrams

Introduction

Flower consists of many parts. Each part has different structure and function. In this activity, you will be dissecting, or taking apart, a flower piece by piece.

Points for pre-lab discussion
Q.1. Have you ever looked at a flower in detail?
Q.2. What parts do flowers consist of?
Q.3. Are all flowers alike?

Procedure

- Label each of the paper plates with one plant part (Stem, Leaf, Sepal, Petal, Stamen and Carpel).
- Draw lines on each paper plate to divide it into four sections.
- Label each section on each plate with a name of one of the four flowering plant.
- Carefully look at each of the flowering plants by using magnifying lens.
- Choose one of your flowering plants and start your plant dissection. Use your hands, scissors, or forceps and carefully take apart your plant.
- Once you have removed one part of the plant, try to identify it, and place it on the corresponding plate.
- Next, repeat the dissection with the remaining three flowering plants.
- Compare the plant parts on each paper plate.
- Draw the diagrams of four flowers that you have dissected.

Flower 1	Flower 2

Studyoesofficial

CamScanner

Flower 3	Flower 4

Post Lab Discussion

Q.1. How many whorls of parts in a flower?

Q.2. What do you notice about petals of different flowers?

Q.3. What is the function of sepals?

Q.4. How petals help in pollination?

Q.5. Where pollen grains are produced in flowers?

Q.6. Which part will make seeds later?

Q.7. Which part of flower can make fruit?

Q.8. Can you plan similar experiment to dissect and study seeds?

Studybeesofficial

CamScanner

Activity 18: To study the germination of seeds

Materials required	Main skills practised
Soaked seeds (Gram and Beans) Washed sand Beakers/ small pots	Following instructions Observing process Interpreting results Making comparisons Drawing diagrams

Introduction

Seed germination is the growth of seedling from seed. In this activity, you will be germinating seeds.

Points for pre-lab discussion
Q.1. Have you ever observed the growth of a seed?
Q.2. What are cotyledons of seed?
Q.3. Do all seeds grow alike?

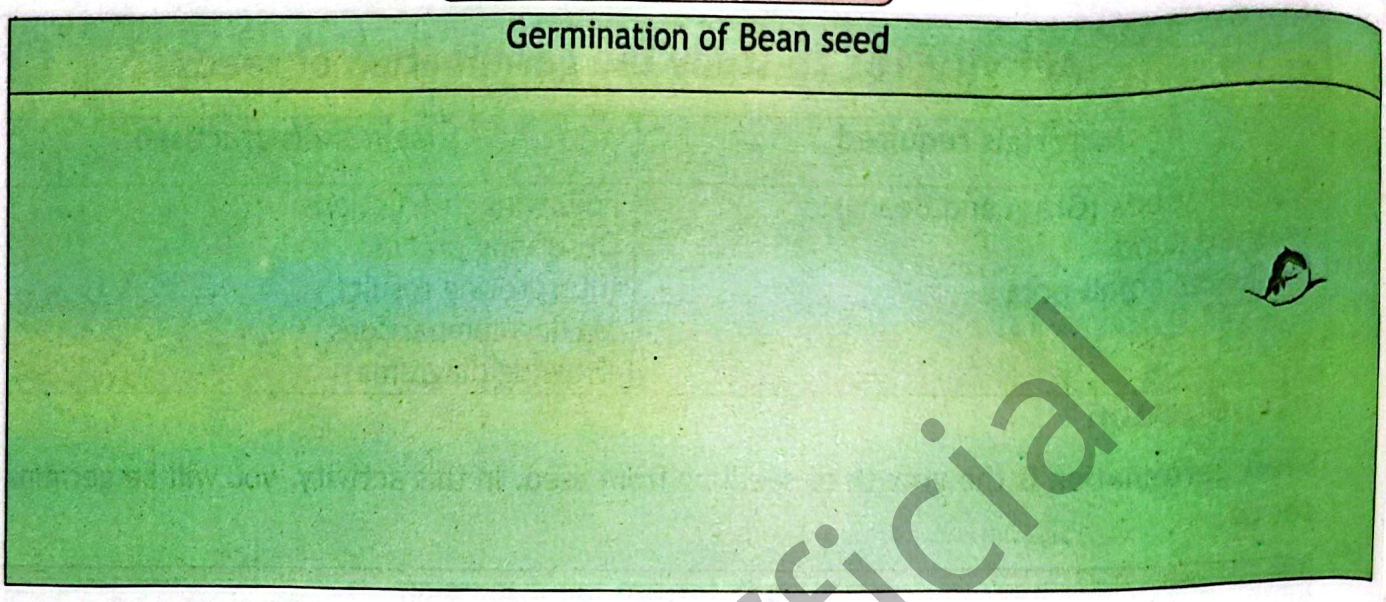
Procedure

- Soak the gram and bean seeds overnight.
- Rinse those seeds with clean water.
- Take two beakers and label them 1 and 2.
- Fill the beakers with washed sand.
- Sow more than 10 seeds in each beaker i.e., gram seeds in beaker 1 and bean seeds in beaker 2.
- Keep the sand moist.
- Take out one seed/ seedling daily for 10 days.
- Observe the growth of seeds/seedlings daily.
- Compare the germination of seeds in both beakers.
- Draw the diagrams of seed germination steps.

Germination of Gram seeds

Studybeesofficial

CamScanner



Post Lab Discussion

Q.1. What is the difference between germination of gram seeds and bean seeds?

Q.2. Which part forms root?

Q.3. Which part forms shoot?

Q.4. What are the two types of germination?

Q.5. Why cotyledons come out of soil during epigeal germination?

Q.6. Why cotyledons stay inside soil during hypogeal germination?

Studybeesofficial

CamScanner

Activity 19: To study continuous and discontinuous variation

Materials required	Main skills practised
Measuring tape Graph paper	Performing experiment Interpreting results Making graphs

Introduction

Variation is any difference between the individuals in groups of organisms. In this activity, you will be analysing different types of variations.

Points for pre-lab discussion
Q.1. What are different types of variations in students of your class?
Q.2. Are these variations depending on genes only?
Q.3. Which variations are affected by environment?

Procedure

- Stand against a wall and extend your arms out to the sides.
- Have a friend mark the spots on the wall where the tips of your middle fingers reach.
- Measure the distance between these two points with measuring tape.
- Record the arm span of all class students in the given table.

Arm span length	Number of students
4'9"	
5'	
5'1"	
5'2"	
5'3"	
5'4"	
5'5"	
5'6"	
5'7"	
5'8"	
5'9"	
6'	
6'1"	
6'2"	

- Make a graph of this data by taking number of students on y-axis and arm span length on x-axis.

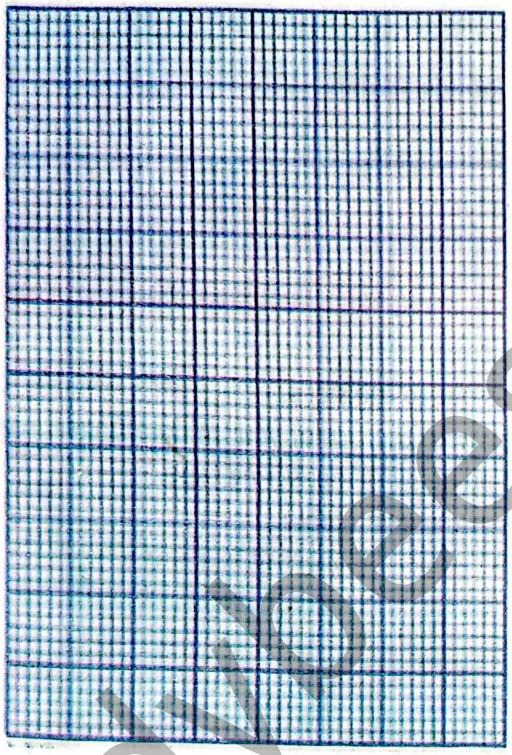
Studybeesofficial

CamScanner

- Ask all students to roll their tongues.
- Count the number of students who can roll their tongues and vice versa.
- Record your observations in following table.

Number of tongue rollers (TR)	Number of non-tongue rollers (NTR)

- Make a graph of this data by taking number of students on y-axis and two groups of students (TR and NTR) on x-axis.



Arm span length graph

Tongue Roller vs non-tongue roller graph

Post Lab Discussion

Q.1. What is the difference between the two graphs you made?

Q.2. Which variation can make a continuous line graph?

Q.3. Which variation (arm span length vs tongue rolling) is not affected by environment?

Q.4. Do you know that your arm span measurement is related with your height?

Q.5. Identify at least 5 continuous and discontinuous variations in humans?

Study Resource official

CamScanner

Activity 20: To study the ecosystem by using sampling technique

Materials required	Main skills practised
Quadrat Open space of ecosystem e.g., school playing ground, park, backyard etc.	Following instructions Planning experiments Observing organisms Recording results Making conclusions

Introduction

Quadrats are used to survey plants and slow-moving animals. They can be used to estimate the number of different species present in an area. You will perform an activity to study biodiversity richness in an ecosystem by using quadrat sampling technique.

Points for pre-lab discussion
Q.1. What is ecosystem?
Q.2. What are biotic factors of ecosystem?
Q.3. What is biodiversity?
Q.4. Why the richness of biodiversity is important in an ecosystem?

Procedure

- Visit a large open area like school playground or park.
- Throw a quadrat gently in the open area.
- Identify plants and animals in the sample area. You can use common or local names as well.
- Count the number of organisms in that area.
- Record your results in the following table.
- Roll or throw your quadrat to two other regions of the same area.

Serial #	Name of plants	Number of plants			Name of animals	Number of animals		
		I	II	III		I	II	III
1	Grass				Ants			
2	Dandelion				Grass hopper			
3	Yellow oxalis				Earthworm			
4					Termite			
5					Snail			
6								

Studybeesofficial

CamScanner

Post Lab Discussion

Q.1. Which plants and animals are more common in the ecosystem you studied?

Q.2. A quadrat can be of any shape or size. What is the shape and size of quadrat you used in this activity?

Q.3. A quadrat can be of metal, PVC, plastic, cardboard, paper etc. Which material is used to make your quadrat?

Q.4. Is it easy to estimate the biodiversity richness in a large ecosystem?

Q.5. Can you use this sampling technique to study all types of ecosystem? Give reasons to support your answer.

Q.6. Do you know that you can identify plants and animals by using Google lens? Ask your teacher to demonstrate.

Studybeesofficial

CamScanner

قومی ترانہ

پاک سَر زمین شاد باد! کشورِ حسین شاد باد!
تو نِشانِ عزمِ عالی شان ارضِ پاکستان
مرکزِ یقین شاد باد!

پاک سر زمین کا نِظام قُوّتِ اُخوّتِ عوام
قوم ، مُلک ، سلطنت پائندہ تابندہ باد!
شاد باد منزلِ مُراد!

پرچَمِ سِتارہ و ہِلال رہبرِ ترقی و کمال
ترجُمانِ ماضی، شانِ حال جانِ اِستقبال
سایۂ خُدائے ذُوالجلال!

Studybeesofficial

National Book Foundation
as
Federal Textbook Board
Islamabad

CamScanner